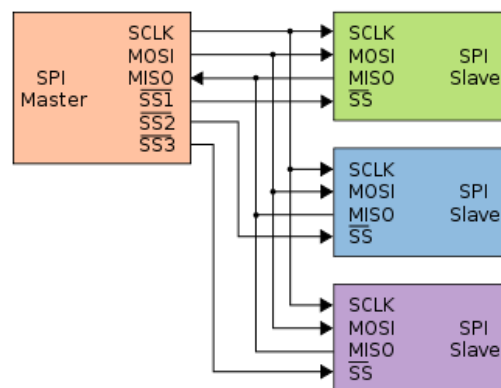


- **ADC Calculation**

- Unipolar, The number of bits needed to achieve resolution of 0.1v. Suppose $V_{ref} = 0v$, $V_{fs} = 4v$, $G=1$.
- Unipolar, What is the resolution of the ADC if we use 8 bits Suppose $V_{ref} = 0v$, $V_{fs} = 4v$, $G= 1$.
- Bipolar, The number of bits needed to achieve resolution of 0.1v. Suppose $V_{ref} = 0v$, $V_{fs} = +/-4v$, $G=1$.
- Bipolar, What is the resolution of the ADC if we use 8 bits Suppose $V_{ref} = 0v$, $V_{fs} = +/-4v$, $G= 1$.

- **Serial Peripheral Interface (SPI)**

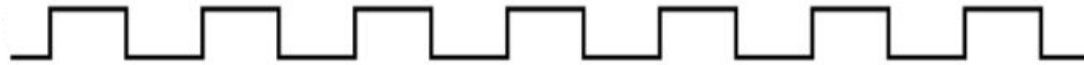
Shown below is a digital temperature sensor peripheral selected by SS1, a digital humidity sensor peripheral selected by SS2 and a digital ambient light sensor peripheral selected by SS3 that use an SPI interface to measure temperature, humidity and ambient light of a refrigeration compartment.



Analyze the constraints described below and fill in the appropriate logic levels of the signals (i.e. 1s and 0s) at each half cycle of the clock.

- Just before the first clock pulse, the master controller selects the temperature sensor which then transmits (MSB first, and centered on the falling edge) a 2-bit value of 2 to the master controller.
- Immediately after the temperature sensor message is completed, the master controller selects the humidity sensor which then transmits (MSB first, and centered on the falling edge) a 3-bit value of 6 to the master controller.
- Immediately after the humidity sensor message is completed, the master controller selects the ambient light sensor which then transmits (MSB first, and centered on the falling edge) a 2-bit value of 1 to the master controller.

D. Immediately after the ambient light sensor message is completed, the master controller does not select any peripheral device



MOSI	X	X	X	X	X	X	X	X	X	X	X	X	X	X
MISO	1	1	0	0	1	1	1	1	0	0	0	0	1	1
$\overline{SS1}$	0	0	0	0	1	1	1	1	1	1	1	1	1	1
$\overline{SS2}$	1	1	1	1	0	0	0	0	0	0	1	1	1	1
$\overline{SS3}$	1	1	1	1	1	1	1	1	1	1	0	0	0	0